AI ASSISTED CODING

Roll no: 2303A52454
B-34

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Lab Objectives

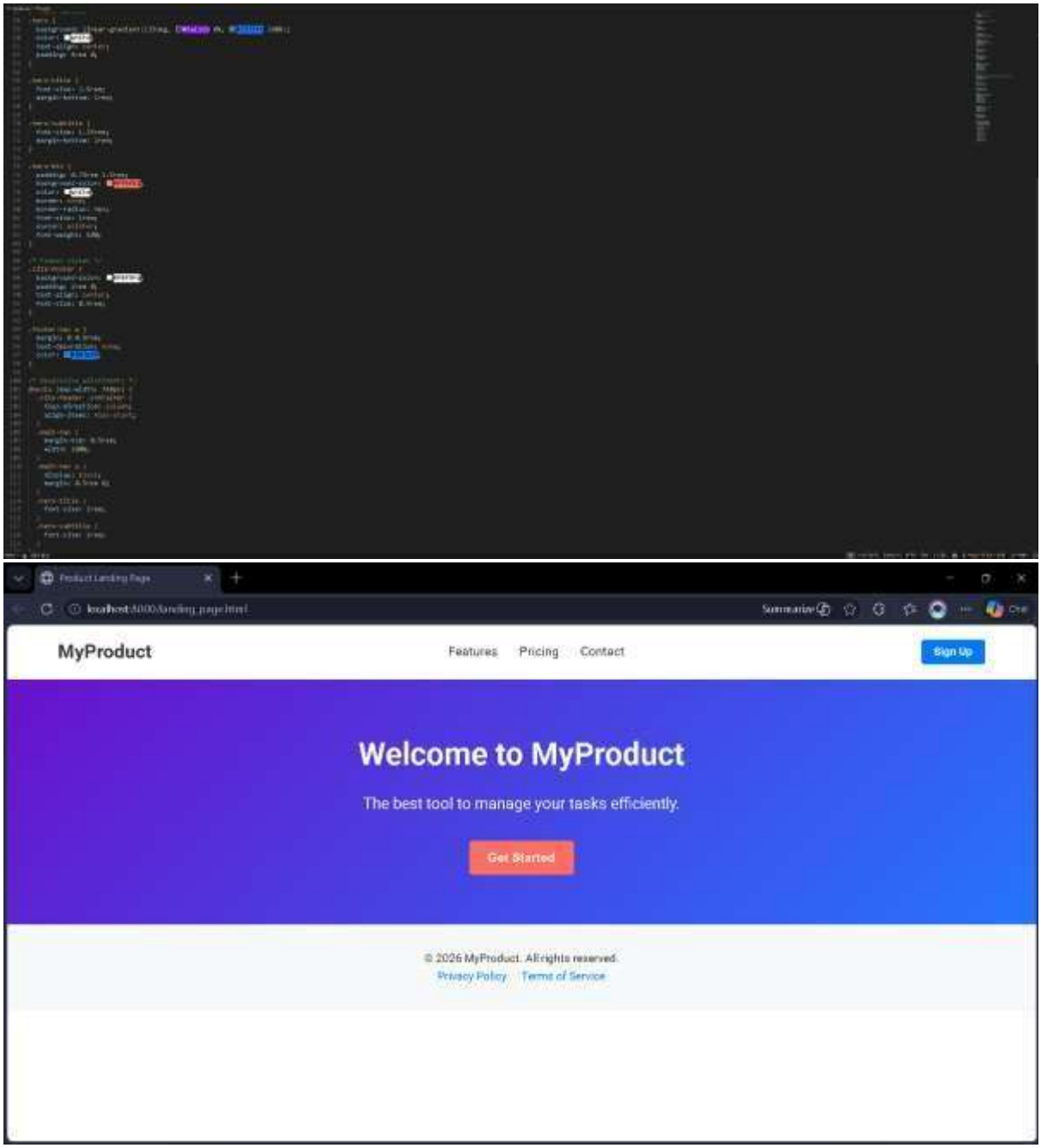
- *Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.*
- *Apply responsive, accessible, and interactive design principles.*
- *Create practical UI components for real-world web applications.*
- *Use AI to optimize layout, UX, and performance.*

Task Description 1 – (Landing Page Design)

- *Create a simple landing page for a product or service. Include a header, hero section, and footer.*

Requirements:

- *Use HTML and CSS to structure the page.*
- *Use AI to suggest layout alignment and typography.*
- *Ensure the page is responsive across devices.*



Task Description 2 – (Theme Toggle Feature)

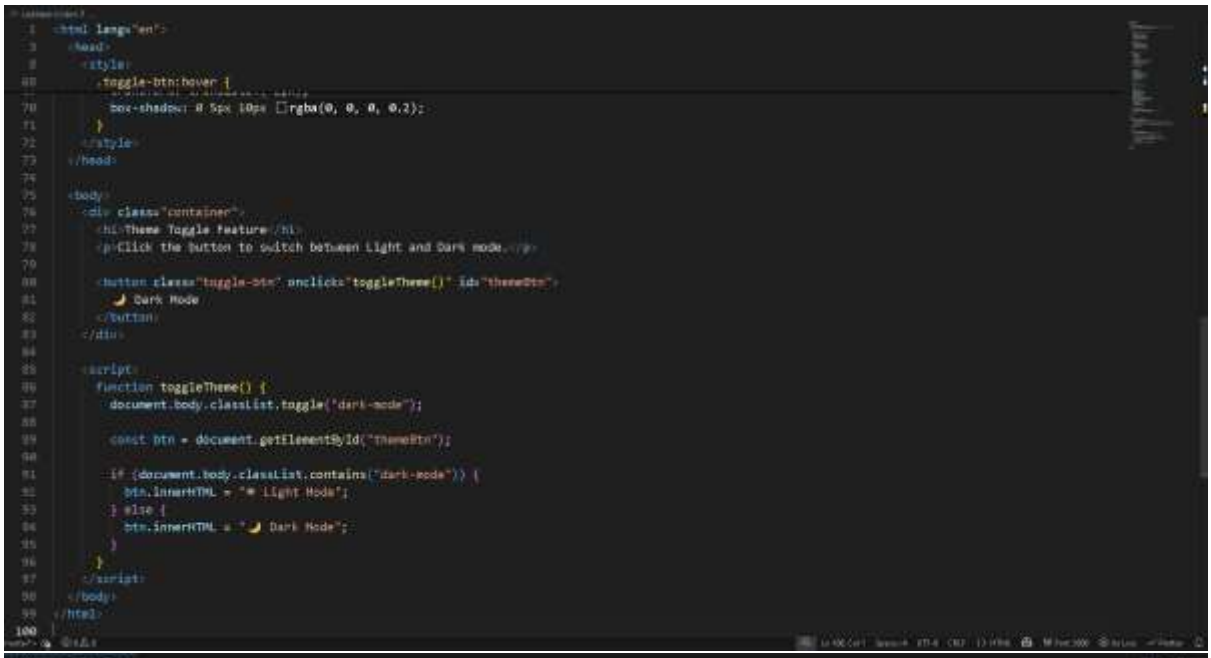
• Provide a button to switch between light and dark themes..

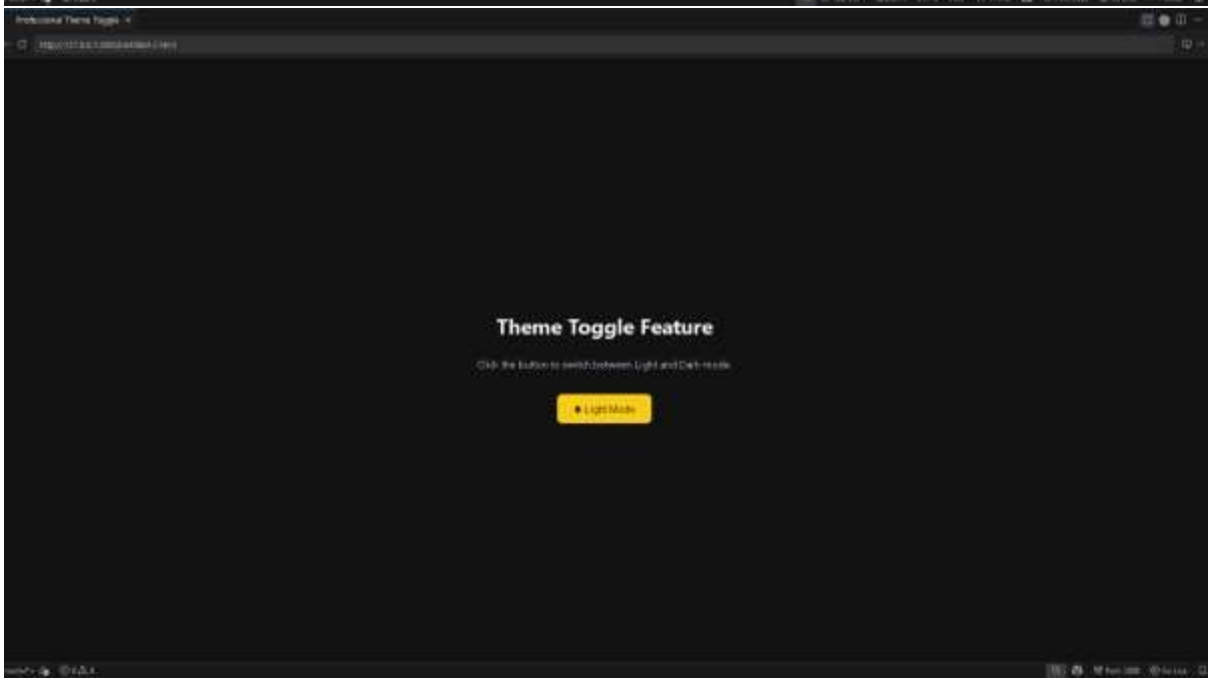
Requirements:

• Use JavaScript to toggle CSS classes.

• Use AI assistance to manage color contrast and accessibility.

• Ensure theme preference applies instantly on click.





Task Description 3 – (User Feedback Form)

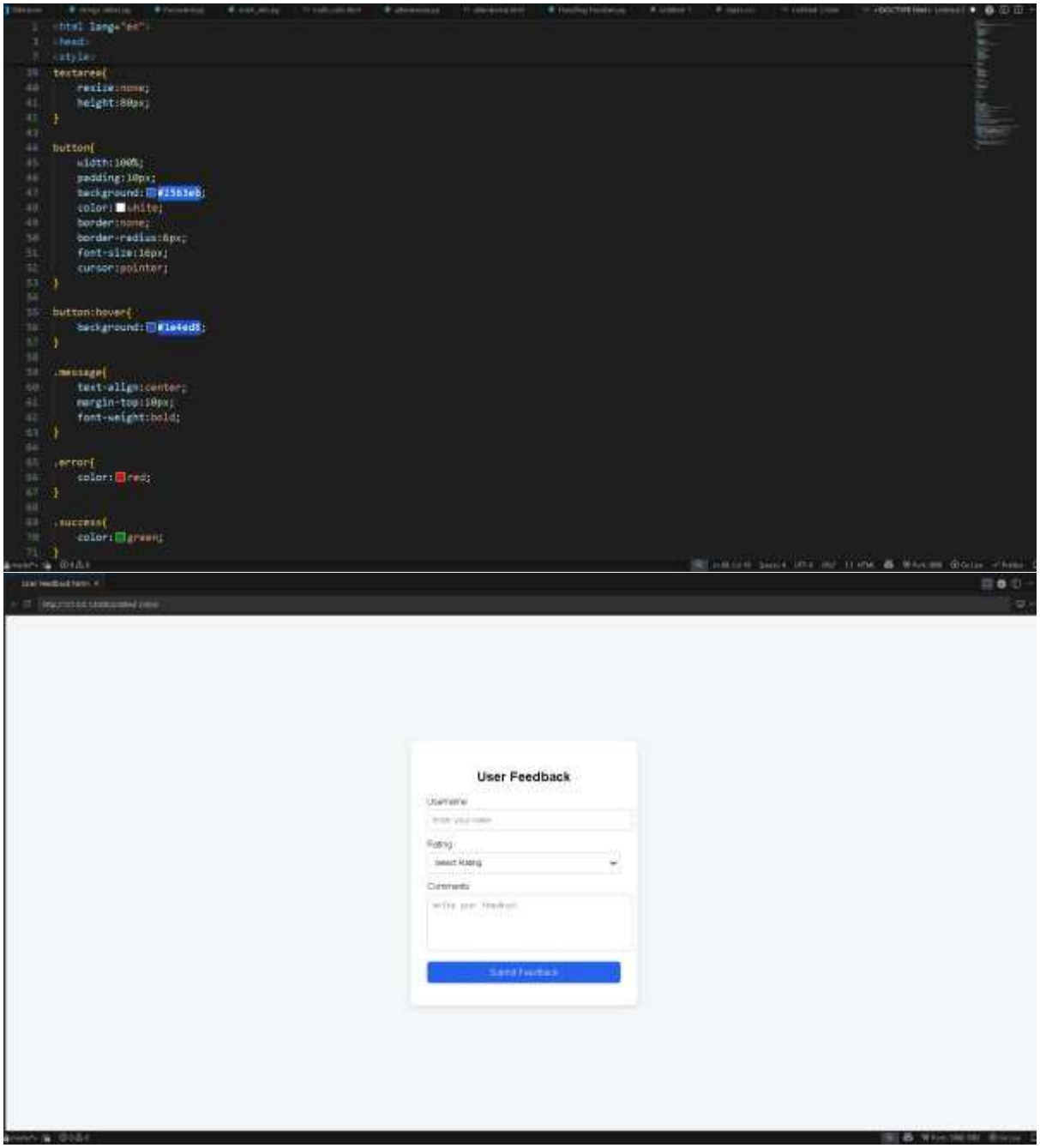
• Create a feedback form with fields: Username, Rating, Comments.

Requirements:

• Use JavaScript to validate required fields.

• Display success or error messages dynamically.

• Use AI to improve form usability and layout.



Task Description - 4 (Image Gallery with JavaScript)

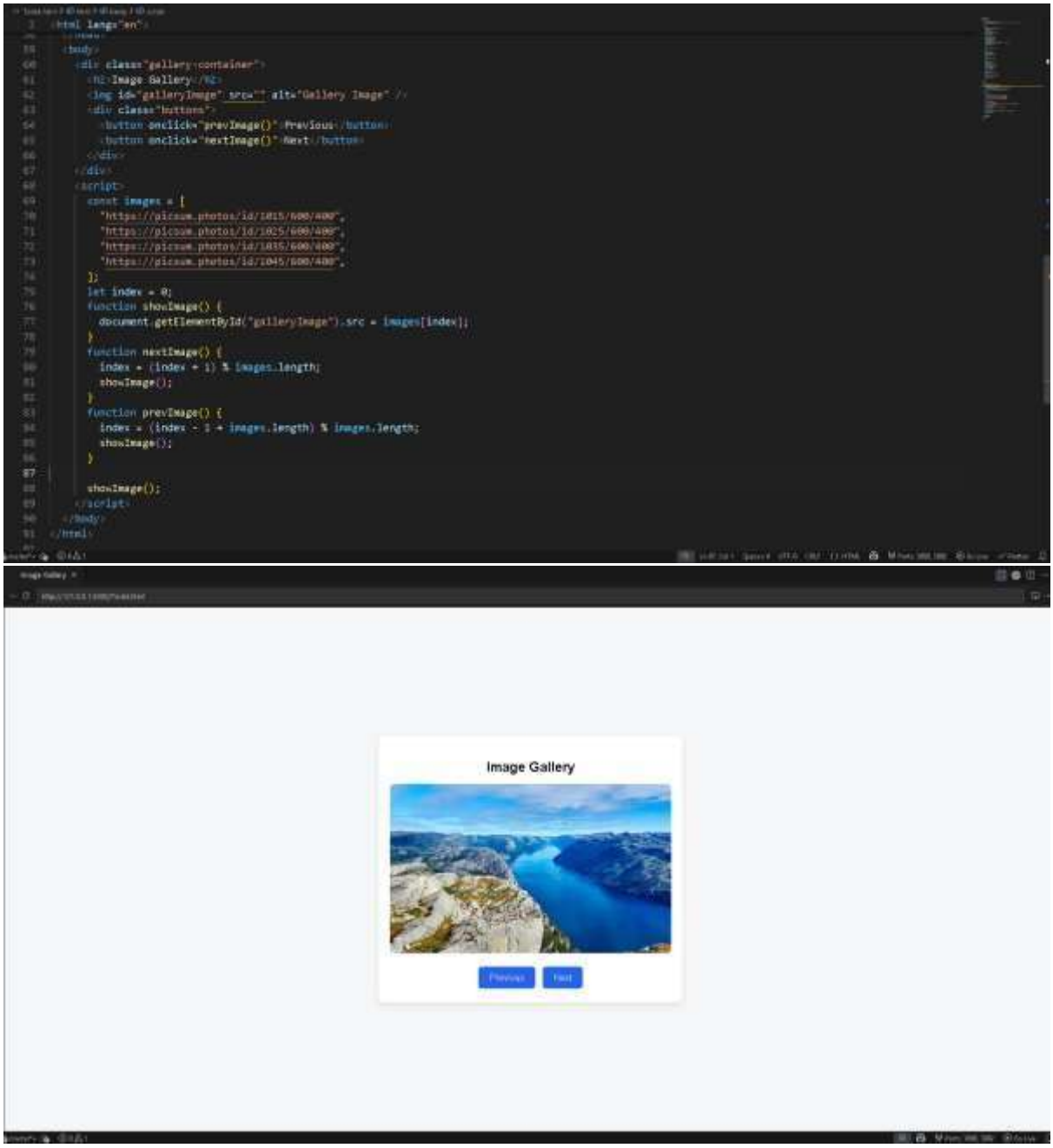
• Task: Display a small image gallery on a web page.

Requirements:

• Use JavaScript to dynamically load images.

• Add next/previous navigation buttons.

• Use AI to optimize layout responsiveness.



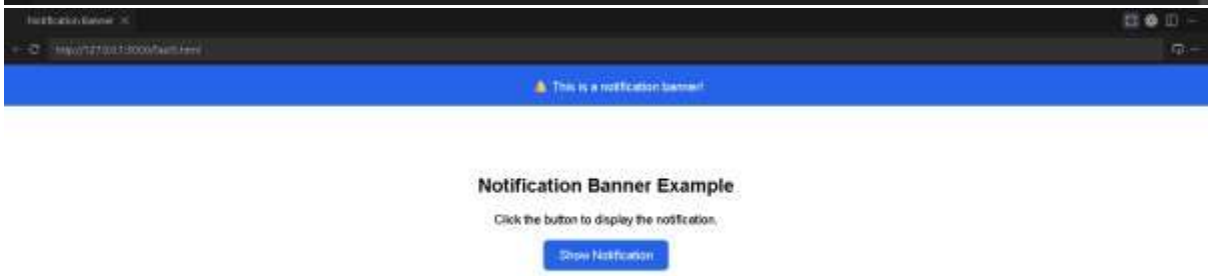
Task Description - 5: (Notification Banner)

• Display a notification banner at the top of the page.

Requirements:

• Use JavaScript to show and hide the banner.

• Style the banner using CSS.

• Use AI to ensure smooth transitions and clean UI behavior.

```html
    <script>
      function showBanner() {
        document.getElementById("notificationBanner").classList.add("show");
      }
      function hideBanner() {
        document.getElementById("notificationBanner").classList.remove("show");
      }
    </script>
  </body>
</html>
```